

Pressure Build-up Tests in Heterogeneous Radial Reservoirs

Theory, a synthetic dataset, and a neural inverse model

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Abstract

A pressure build-up test shuts in a producing well and records how the bottomhole pressure recovers. Its purpose is an inverse calculation: from one pressure curve, infer the permeability around the well, the skin, the wellbore storage and the distance to the reservoir boundary. This report first develops the theory of build-up testing (flow regimes, superposition, the Horner and derivative methods, and material balance with its dependence on the initial pressure p_i), and then describes a set of numerical experiments: a validated finite-volume simulator, a dataset of 20 000 heterogeneous closed reservoirs, a check of what a build-up actually measures, a model-based inversion, and a neural inverse model (a Fourier neural operator along log time) that returns the radial permeability profile, the storage and the boundary distance with calibrated uncertainties. With the initial pressure known and a precise gauge, the model recovers the boundary distance to 1–2 %; without p_i the error is about 20 %, for a physical reason discussed in Section 2.5.

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1 Introduction: what a build-up test is for

A *pressure transient test* perturbs a well’s rate and records the bottomhole pressure response. Because the pressure disturbance travels into the reservoir by diffusion, the response at early times reflects the well and its immediate surroundings, and the response at later times reflects regions progressively farther away. Reading the response as a function of time is therefore reading the reservoir as a function of distance.

Two elementary tests exist. In a *drawdown* the well is opened and produced at (ideally) constant rate while the flowing pressure $p_{wf}(t)$ falls. In a *build-up* a well that has produced for a time t_p is shut in and the shut-in pressure $p_{ws}(\Delta t)$ recovers towards the reservoir pressure. Build-ups are the more common diagnostic test in practice: a zero rate is exactly controlled (a constant nonzero rate rarely is), and the well is tested without the rate fluctuations and gauge disturbances of flowing conditions. The price is lost production during the shut-in, which limits how long a test can last.

A build-up interpretation delivers:

- the *permeability–thickness* kh of the formation, from the radial-flow regime;
- the *skin* S , the extra pressure drop near the well (damage or stimulation);
- the *wellbore storage* C , the compressibility of the fluid column in the well;
- *boundaries*: faults, the closed outer limit of the drainage area and its distance;
- the *average reservoir pressure* $\bar{p}$, needed for material balance and reserves.

The classical workflow identifies flow regimes on a log–log diagnostic plot (pressure change and its logarithmic derivative), reads parameters from specialized plots (Horner, semilog, Cartesian), and refines them by matching an analytical or numerical model to the data. The rest of the report builds this theory (Section 2) and then applies it to synthetic tests in heterogeneous reservoirs, where the classical assumptions are strained (Sections 3–6).

2 Theory of build-up testing

2.1 The radial diffusivity equation

For single-phase flow of a slightly compressible fluid (constant viscosity μ and total compressibility c_t) towards a vertical well of radius r_w in a horizontal layer of thickness h and porosity ϕ , mass conservation and Darcy’s law give, for a permeability that varies with radius only,

$$\phi\mu c_t \frac{\partial p}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k(r) \frac{\partial p}{\partial r} \right), \quad r_w \leq r \leq r_e. \quad (1)$$

With a reference permeability k_{ref} and a reference rate q , the dimensionless variables

$$r_D = \frac{r}{r_w}, \quad t_D = \frac{k_{\text{ref}} t}{\phi \mu c_t r_w^2}, \quad k_D = \frac{k}{k_{\text{ref}}}, \quad p_D = \frac{2\pi k_{\text{ref}} h (p_i - p)}{q B \mu}, \quad C_D = \frac{C}{2\pi \phi c_t h r_w^2} \quad (2)$$

(B is the formation volume factor, C the wellbore storage coefficient) turn (1) into

$$\frac{\partial p_D}{\partial t_D} = \frac{1}{r_D} \frac{\partial}{\partial r_D} \left(r_D k_D \frac{\partial p_D}{\partial r_D} \right), \quad 1 \leq r_D \leq r_{eD}. \quad (3)$$

p_D is the pressure *drop* below the initial pressure p_i , per unit rate: it starts at zero and grows as the well produces. Time scales with $k/(\phi \mu c_t)$, the hydraulic diffusivity; for example, with $k = 100$ md, $\phi = 0.2$, $\mu = 1$ cP, $c_t = 10^{-9}$ Pa $^{-1}$ and $r_w = 0.1$ m, one unit of t_D is 0.02 s, so $t_D = 10^6$ is about 5.6 hours.

2.2 Boundary conditions: rate, storage, skin, closed boundary

Well. The well produces at a prescribed surface rate $q_D(t_D)$ (1 while producing, 0 after shut-in). The fluid in the wellbore is compressible, so right after a rate change the sandface rate q_{sf} differs from the surface rate; the difference is the *wellbore storage*,

$$C_D \frac{dp_{wD}}{dt_D} + q_{sfD} = q_D(t_D), \quad q_{sfD} = -k_D r_D \frac{\partial p_D}{\partial r_D} \Big|_{r_D=1}. \quad (4)$$

Skin. Drilling damage, perforations or stimulation change the permeability close to the well. The classical *thin skin* adds a pressure drop proportional to the sandface rate, $p_{wD} = p_D(1) + S q_{sfD}$. A *finite* altered zone of permeability k_s out to r_s in a formation of permeability k is equivalent in steady state to the skin of Hawkins,

$$S = \left(\frac{k}{k_s} - 1 \right) \ln \frac{r_s}{r_w}, \quad (5)$$

positive for damage ($k_s < k$) and negative for stimulation.

Outer boundary. A closed (no-flow) circular boundary at r_e : $\partial p_D / \partial r_D = 0$ at $r_D = r_{eD}$. The initial condition is $p_D = 0$ everywhere.

2.3 Flow regimes of a drawdown

For a homogeneous reservoir ($k_D = 1$) producing at unit rate, the well pressure goes through three regimes.

Wellbore storage. At first the well produces fluid from the wellbore itself, so $q_{sfD} \approx 0$ and (4) gives $p_{wD} = t_D / C_D$: a straight line of *unit slope* on a log-log plot. Storage effects fade after about $t_D \approx (60 + 3.5 S) C_D$ (Ramey's rule of thumb).

Infinite-acting radial flow. Before the boundary is felt, the reservoir behaves as if infinite. The line-source solution $p_{wD} = \frac{1}{2}E_1(1/(4t_D))$ becomes, for $t_D \gtrsim 25$,

$$p_{wD} = \frac{1}{2}(\ln t_D + 0.80907) + S, \quad (6)$$

a straight line in $\ln t$ of slope $1/2$ (for permeability k : slope $1/(2k_D)$ and $\ln(k_D t_D)$). In field units this is the familiar semilog slope $m = 162.6 qB\mu/(kh)$ psi per log cycle. The pressure front has then reached roughly the *radius of investigation* $r_{\text{inv},D} \approx 2\sqrt{k_D t_D}$, so the slope at time t reflects the permeability around that radius.

Pseudo-steady state. Once the whole closed reservoir is drained, the pressure drops at the same rate everywhere, and

$$p_{wD} = \frac{2t_D}{r_{eD}^2 - 1} + \ln r_{eD} - \frac{3}{4} + S. \quad (7)$$

The rate of decline $dp_{wD}/dt_D = 2/(r_{eD}^2 - 1)$ is set by the pore volume alone; this is the basis of the *reservoir limit test*. The boundary starts to be felt around $t_D \approx r_{eD}^2/(4k_D)$.

2.4 Superposition and the build-up

Equation (3) and its boundary conditions are linear in p_D , for any $k_D(r)$ and constant C_D , S . A shut-in at t_p is therefore the sum of the original production continued forever and an injection of the same rate starting at t_p . With $p_{dd}(t)$ the unit-rate drawdown,

$$p_{wD}(t_p + \Delta t) = p_{dd}(t_p + \Delta t) - p_{dd}(\Delta t), \quad \Delta p_{bu}(\Delta t) = p_{ws} - p_{wf} = p_{dd}(t_p) - p_{dd}(t_p + \Delta t) + p_{dd}(\Delta t). \quad (8)$$

This is exact, including storage (the afterflow after shut-in) and heterogeneity.

Horner method. If both $p_{dd}(t_p + \Delta t)$ and $p_{dd}(\Delta t)$ are in radial flow, (6) and (8) give

$$p_{wD}(t_p + \Delta t) = \frac{1}{2} \ln \frac{t_p + \Delta t}{\Delta t}, \quad (9)$$

a straight line of slope $1/(2k_D)$ against the logarithm of the *Horner time* $(t_p + \Delta t)/\Delta t$. Extrapolating to Horner time 1 ($\Delta t \rightarrow \infty$) gives the pressure p^* , which equals p_i for an infinite-acting reservoir. The skin follows from the pressure change at a reference time on the straight line.

Agarwal time and the Bourdet derivative. Plotting a build-up against the Agarwal equivalent time $\Delta t_e = t_p \Delta t / (t_p + \Delta t)$ makes it resemble a drawdown, so drawdown type curves apply. The *Bourdet derivative* $d\Delta p_{bu}/d \ln \Delta t_e$, computed with a smoothing window L in $\ln \Delta t_e$, is the key diagnostic: storage gives a unit slope that coincides with the pressure curve, skin appears as a hump, radial flow as a horizontal plateau at $1/(2k_D)$, and boundaries as departures from the plateau (a closed boundary makes the build-up derivative fall as the pressure stabilizes). Figure 1(a,b) shows these regimes for a homogeneous reservoir, computed with the simulator of Section 3.1: the fitted Horner slope is 0.503 per natural-log unit (theory 0.5) and p^* is within 0.005 of p_i .

If production lasted long enough for the boundary to be felt before shut-in, $p_{dd}(t_p + \Delta t)$ is no longer semilog, the Horner line is distorted and $p^* \neq p_i$; the Matthews–Brons–Hazebroek (MBH) method relates $p^* - \bar{p}$ to the dimensionless production time $t_{pDA} = kt_p/(\phi\mu c_t A)$ of the drainage area A .

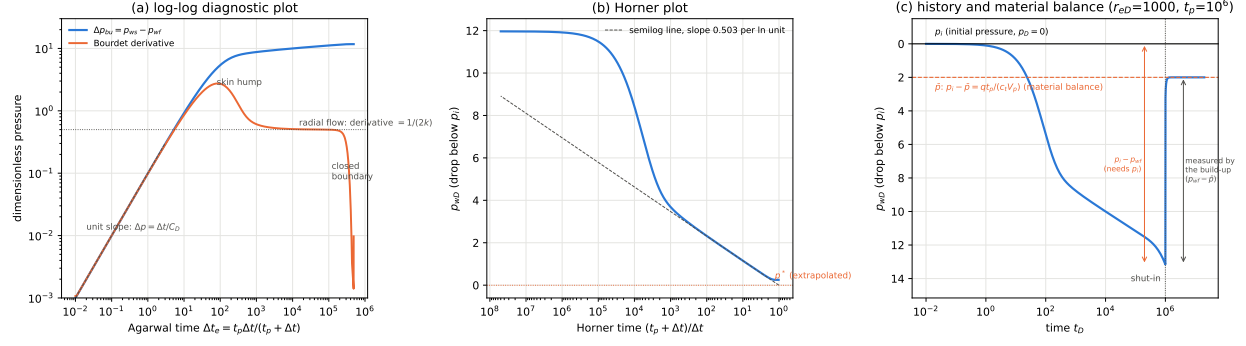


Figure 1: Build-up theory in a homogeneous closed reservoir ($k_D = 1$, $C_D = 10$, $S = 5$), from the simulator. (a) Diagnostic plot for $r_{eD} = 2000$, $t_p = 5 \times 10^5$ (infinite-acting at shut-in): unit-slope storage, skin hump, radial-flow plateau at $1/2$, closed-boundary drop. (b) Horner plot of the same test: semilog slope 0.503, extrapolated $p^* = p_i$. (c) Complete history of a longer-producing test ($r_{eD} = 1000$, $t_p = 10^6$): the build-up stabilizes at the average pressure $\bar{p}$ given by material balance; the gauge measures only the rise $p_{ws} - p_{wf}$, whereas $p_i - p_{wf}$ and $p_i - \bar{p}$ require the initial pressure.

2.5 Material balance and the role of the initial pressure

In a closed reservoir, every barrel produced comes from the expansion of the fluid and rock in the drainage volume and from the wellbore. After a production period t_p at rate q and a shut-in long enough for the pressure to equalize at $\bar{p}$,

$$qB t_p = c_t V_p (p_i - \bar{p}) + C (p_i - \bar{p}), \quad V_p = \pi \phi h (r_e^2 - r_w^2), \quad (10)$$

which in the variables (2) reads

$$\bar{p}_D = \frac{t_{pD}}{(r_{eD}^2 - 1)/2 + C_D}. \quad (11)$$

This is independent of the permeability: the *volume* of the reservoir, and hence r_{eD} , follows directly from the depletion $p_i - \bar{p}$ and the produced volume. The test simulator satisfies (11) to rounding error, and on stabilized synthetic build-ups it returns r_{eD} to 3×10^{-7} (Section 4).

The catch is what the gauge measures (Figure 1c). A build-up records $p_{ws}(\Delta t)$, and its interpretation uses the *rise* $p_{ws} - p_{wf}$. Material balance needs the *depletion* $p_i - \bar{p} = (p_i - p_{wf}) - (\bar{p} - p_{wf})$: the build-up supplies the second term, but the first term needs p_i . Three situations follow.

- **p_i known.** Typical for a new or exploration well: the initial pressure is measured by a formation tester, a drill-stem test or a static survey before production. Then $p_i - p_{wf}$ is known and the stabilized build-up gives r_{eD} by (11). Even before stabilization, the apparent size $r_{app}(\Delta t) = \sqrt{2t_p / (p_{wf} - \Delta p_{bu}(\Delta t)) + 1}$ is a lower bound that converges to r_{eD} ; the neural model of Section 6 uses it as an input.
- **p_i unknown, single build-up.** Only differences within the build-up are available. The boundary still shows up through its *timing*, $t_D \approx r_{eD}^2 / (4k_D)$, but timing constrains r_{eD}^2 / k , not r_{eD} : a larger reservoir with more permeable far field looks the same. The boundary distance is then only as good as the knowledge of the far-field permeability, which the build-up constrains least. This is the same ambiguity that makes the MBH method require k and the shape of the drainage area.

- **p_i unknown, more data.** Volume can still be measured without p_i from pressure *changes over produced volume*: the slope of the flowing pressure in pseudo-steady state ((7), a reservoir limit test), or the difference between the stabilized pressures of two build-ups separated by a known production (the two flow periods of a drill-stem test).

The average pressure $\bar{p}$ itself is usually an *output* of the build-up (via p^* and MBH), not an input.

2.6 Heterogeneous reservoirs: what the derivative measures

When k varies with radius, there is no single radial-flow plateau. At time t the derivative reflects the permeability around the radius of investigation $2\sqrt{kt}$, averaged over a region of roughly half a decade in radius. Radial resistances add in series, so the natural average of k over a ring $[a, b]$ is the *log-harmonic mean*

$$\bar{k}_{[a,b]} = \ln \frac{b}{a} \bigg/ \int_a^b \frac{dr}{r k(r)}, \quad (12)$$

and any near-well region of different permeability acts as a skin relative to whatever k is read farther out. A build-up therefore constrains a smoothed radial permeability *profile*, sharply near the well and more loosely farther out; a single pair (k, S) is a summary whose value depends on the time window chosen. Section 4 quantifies this on the synthetic data.

3 Synthetic build-up data

3.1 Simulator

Equation (3) is solved by vertex-centered finite volumes on a log-spaced grid $r_{D,j} = e^{jD}$, $D = \ln 1000/1023 \approx 0.0068$ (about 850–1190 nodes for $r_{eD} = 300$ –3000). Face fluxes $k_{j+1/2}(p_{j+1} - p_j)/D$ are exact for steady radial flow; $k_{j+1/2}$ is the harmonic mean of the node values, exact for a permeability jump at the face. The unknowns $(p_{wD}, p_0, \dots, p_{N-1})$ keep the system tridiagonal, including storage (4) and thin skin; time integration is variable-step BDF2, restarted at rate changes. A full run from $t_D = 0$ to 10^7 takes about 0.05 s. Table 1 summarizes the validation.

Check ($k_D = 1$ unless stated)	Result
van Everdingen–Hurst (Laplace, Stehfest), $r_{eD} = 300$, $t_D = 0.1$ – 10^7 , $C_D \in \{0, 100\}$, $S \in \{0, 5\}$	$\leq 3 \times 10^{-4}$ relative
line source, $t_D = 10^3$ – 10^5	$< 10^{-3}$
pseudo-steady state (7)	$< 10^{-4}$
storage unit slope, $C_D = 10^3$	$< 2 \times 10^{-3}$
mass balance, heterogeneous k , shut-in	exact to 10^{-9}
simulated shut-in vs superposition (8), heterogeneous k	$< 10^{-3}$ of the build-up change
radial-flow interpretation (plateau k , semilog S), with storage and skin	k within 1.5 %, S within 0.15

Table 1: Validation of the simulator and the build-up tools.

Because of superposition, one unit-rate drawdown per reservoir, stored at 200 log-spaced times in $[10^{-2}, 10^7]$, generates every build-up; between the stored times the drawdown is interpolated by a cubic spline in $\ln t$ (accuracy 3×10^{-6}).

A unit convention: the solver’s thin skin is scaled with k_{ref} , whereas the skin read from a build-up is relative to the formation permeability, $S_{\text{wt}} = k_D S$.

3.2 Permeability model

Each reservoir has

$$\ln k_D(r) = m + g(\ln r) + \delta \mathbf{1}[r < r_s], \quad \text{clipped to } [-3, 3], \quad (13)$$

with a mean shift $m \sim N(0, 0.3^2)$, a zero-mean Gaussian random field g in $\ln r$ with squared-exponential covariance of standard deviation $\sigma \sim U(0.5, 1)$ and correlation length $\ell \sim U(0.3, 1.5)$, and, in half of the reservoirs, a skin zone of radius $r_s \sim \log U[2, 20]$ and log permeability ratio $\delta = \ln(k_s/k) \sim U(\ln 0.1, \ln 5)$. The field is sampled by circulant embedding (FFT) on the uniform $\ln r$ grid. It is stationary in $\ln r$: its statistics are the same per unit of $\ln r$ from the wellbore outward (measured spread 0.82–0.85 at every radius), so features are larger farther from the well (Figure 2).

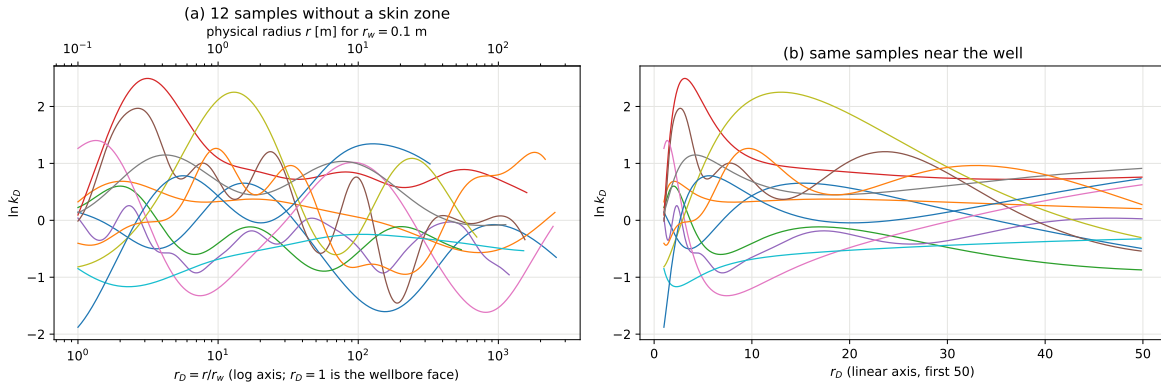


Figure 2: Twelve permeability fields without a skin zone. (a) Log radius, with the physical radius for $r_w = 0.1$ m on top: variation starts at the wellbore face. (b) Linear radius near the well: stationarity in $\ln r$ packs the variation close to the well.

3.3 Dataset

The dataset has 20 000 reservoirs (18 000 training, 1 000 validation, 1 000 test), generated in six minutes on eight cores. Per reservoir: $r_{eD} \sim \log U[300, 3000]$; $C_D = 0$ in 20 % of the reservoirs and $\log U[0.1, 10^3]$ otherwise; the permeability field above; one unit-rate drawdown (stored in float64, since build-ups are small differences of drawdown values). In the training split the radial-flow permeability ranges from 0.07 to 7.2 (median 0.84), and the boundary is felt within $\Delta t_D = 9 \times 10^6$ in 99.9 % of the reservoirs. Figure 3 shows four test reservoirs.

4 What a build-up actually measures

Before training anything, the natural labels were checked against a classical interpretation of each test build-up ($t_p = 10^6$, radial-flow window chosen with the true parameters: after storage, before the boundary; 684 of 1000 test reservoirs have such a window). Figure 4 and Table 2 summarize.

The boundary distance is a clean label; a single (k, S) pair is not, as anticipated in Section 2.6. The inverse problem was therefore posed with the targets

- $\ln \bar{k}$ by (12) over eight rings in r_D : $[1, 2]$, $[2, 5]$, $[5, 10]$, $[10, 30]$, $[30, 100]$, $[100, 300]$, $[300, 1000]$, $[1000, 3000]$, each clipped at r_{eD} and masked when less than a radius ratio 1.5 of it lies inside;

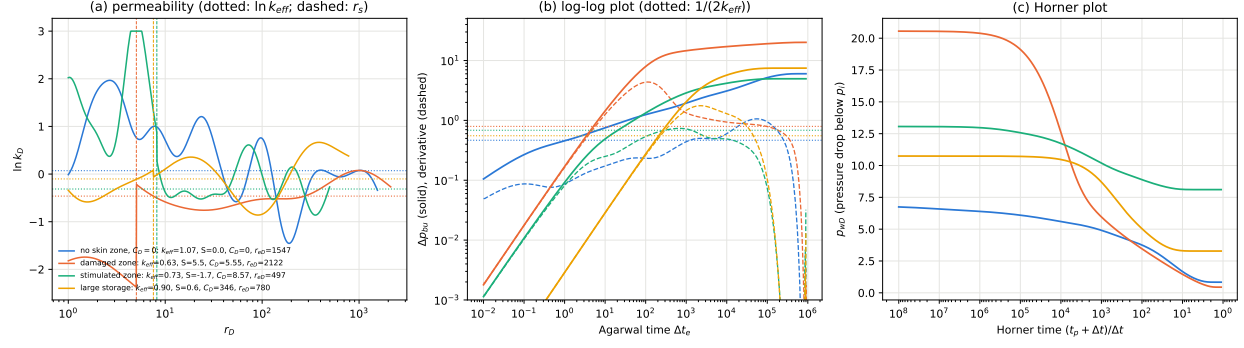


Figure 3: Four test reservoirs after $t_p = 10^6$. (a) $\ln k_D$ (dotted: the whole-reservoir average; dashed: r_s). (b) Diagnostic plot: pressure change (solid) and derivative (dashed); the dotted levels are where the derivative would sit if a single permeability applied. (c) Horner plot. The derivative of the heterogeneous reservoirs wanders instead of forming one plateau.

Label	Check	Result
r_{eD}	material balance (11) from the stabilized build-up (695 tests)	exact to 3×10^{-7}
k (whole-reservoir log-harmonic mean)	vs derivative plateau	median error 23 %, 90th pct. $\sim 2\times$
k over the investigated ring	vs derivative plateau	median error 18 %
S (relative to the whole-reservoir k)	vs semilog skin	median difference 1.1, 90th pct. 6

Table 2: Label check on the test split.

- $\ln C_D$, with a separate no-storage flag;
- $\ln r_{eD}$.

Any skin definition can be computed from the profile afterwards.

5 Model-based inversion

The most direct inverse method fits the simulator to the data. The unknowns are $\ln k$ at 16 control points (linear in $\ln r$ between $r_D = 1$ and 3000), $\ln C_D$ and $\ln r_{eD}$; the misfit is the relative error of the build-up change and of the drawdown at shut-in $p_i - p_{wf}$, for 1 % gauge noise, plus a smoothness penalty and a weak prior on $\ln k$; the minimization is Levenberg–Marquardt-type least squares started from the classical estimates, and the uncertainty is the Gauss–Newton covariance. On test reservoir 486 (damaged zone on a low-permeability near-well region, $S = 9.2$) it recovers $C_D = 0.582$ (true 0.584), $r_{eD} = 1658 \pm 10\%$ (true 1645), and the ring permeabilities to within 0.04 in $\ln k$ for $r_D < 100$ and about 0.25 beyond (Figure 5), in 10–40 s. Fitting the build-up change alone, without $p_i - p_{wf}$, let r_{eD} drift to 1246: the material-balance information enters only through p_i .

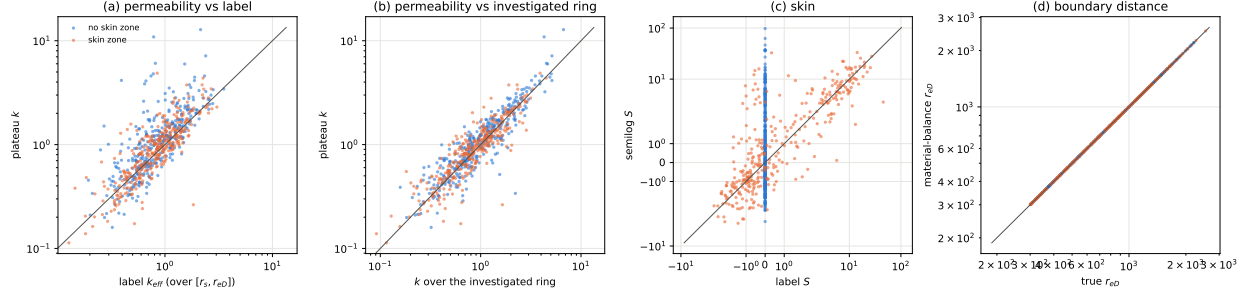


Figure 4: Classical interpretation vs labels. (a, b) The plateau permeability agrees better with the ring actually investigated than with the whole-reservoir average. (c) Reservoirs without a skin zone (label $S = 0$) read semilog skins from -1 to about 100 : near-well heterogeneity acts as skin. (d) Material balance recovers r_{eD} exactly.

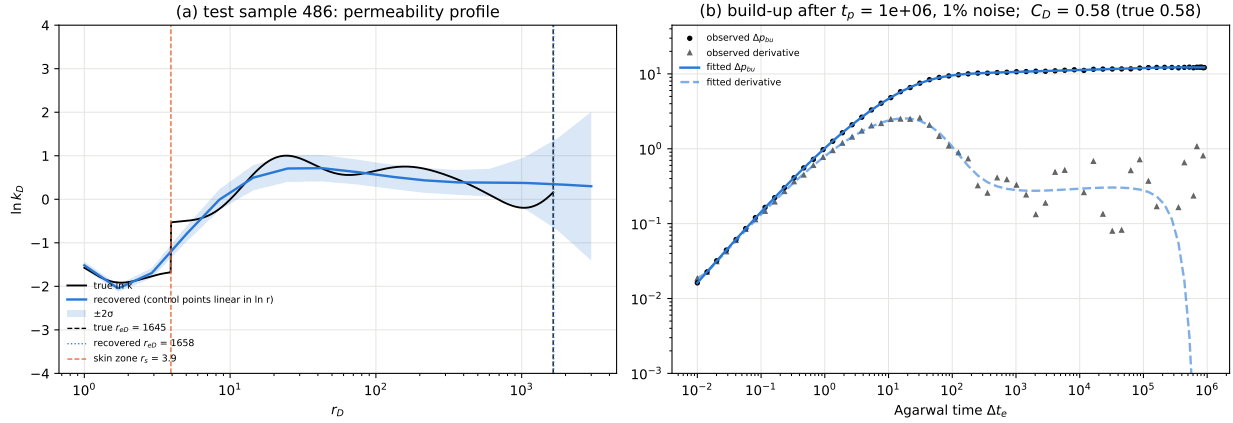


Figure 5: Model-based inversion of test reservoir 486 with 1% noise. (a) True and recovered $\ln k$ with $\pm 2\sigma$; the skin step is smoothed over one control spacing. (b) Observed and fitted build-up.

6 A neural inverse model

The model-based inversion is accurate but takes tens of seconds and needs a starting point. A neural network can learn the same map from data once and then evaluate it in milliseconds.

6.1 Inputs, targets and realism

Every training step forms fresh build-ups from the stored drawdowns by superposition, on a fixed grid of 128 log-spaced shut-in times in $[10^{-2}, 9 \times 10^6]$. The input channels are: $\ln \Delta p_{bu}$; the log-derivative $d \ln \Delta p_{bu} / d \ln \Delta t$; the time coordinate; $\ln t_p$; $\ln(p_i - p_{wf})$ and a p_i -known flag; a validity mask; the apparent material-balance radius $r_{app}(\Delta t)$ (Section 2.5); and the gauge noise level. Realism: $t_p \sim \log U[10^4, 10^6]$; relative gauge noise $\log U[10^{-4}, 10^{-2}]$; p_i unknown in 25% of the examples; half of the tests truncated at $\Delta t_{max} \sim \log U[10^6, 9 \times 10^6]$ (days of shut-in). Production and test length are drawn independently of the reservoir: a test length tied to r_{eD}^2/k would let the model read r_{eD} from the length of the record.

6.2 Architecture and loss

A Fourier neural operator along $\ln \Delta t$ (lift, four spectral layers with 24 modes and width 48, pointwise paths, GELU) extracts features at all time scales; masked mean and max pooling over the valid part of the test feed an MLP that outputs a mean and a log-variance for each of the ten targets and a logit for the no-storage flag (0.48 M parameters). The loss is the β -weighted Gaussian negative log-likelihood ($\beta = 0.5$): with the plain likelihood the gradient of a target’s mean scales with its inverse predicted variance, and r_{eD} , initially assigned a large variance, was never learned (Table 3, first row).

6.3 Development

Table 3 traces the changes that mattered.

Run	Change	r_{eD} median error: all / p_i known / p_i unknown	Test set
base	Gaussian NLL	not learned (prior spread)	short tests
beta	β -NLL	35 / 34 / 39 %	short tests
mb	+ material-balance input r_{app}	31 / 27 / 41 %	short tests
long	realistic production and test lengths	13.6 / 9.5 / 28.5 %	realistic, 1 % noise
noise	noise level randomized and given as input	14.1 / 10.1 / 29.6 %	realistic, 1 % noise
		3.4 / 2.0 / 20.0 %	realistic, 0.1 % noise
		1.7 / 1.1 / 19.6 %	realistic, 0.01 % noise

Table 3: Development of the inverse model (1000 test reservoirs, four build-ups each). “Short tests”: production from $t_p = 10^3$ and truncation down to $\Delta t = 10^2$, where the boundary is often never felt.

Three lessons stand out. First, the material-balance input made the model use p_i : before it, knowing p_i barely helped, and on stabilized tests the model was worse than the material-balance formula. Second, most of the improvement from “mb” to “long” came from the test data, not from training: evaluated on the realistic tests, the “mb” model already reaches 13.8 % (p_i known 9.9 %). Tests that are too short do not contain r_{eD} . Third, a model trained at 1 % noise does not exploit a precise gauge; told the noise level, it does.

6.4 Results

Figure 6 shows the final model on the test set at 0.1 % noise, Figure 7 its dependence on gauge precision, and Table 4 the numbers.

Uncertainty. The predicted standard deviations are calibrated: the empirical coverage follows the nominal level at every level (Figure 8, model trained at 1 % noise), so no post-hoc scaling is used. The noise-aware model is slightly overconfident at 1 % noise (90–94 % coverage of $\pm 2\sigma$).

6.5 When the boundary distance is not determined

The largest r_{eD} errors (the worst 10 %) have two causes (Figure 9, 1 % noise). The first is *drainage before shut-in*, $t_p k / r_{eD}^2$: with p_i known the median error is 33 % when it is below 0.01 and 6.9 %

Gauge noise	0.01 %	0.1 %	1 %
r_{eD} median error, p_i known	1.1 %	2.0 %	10 %
90th percentile	5.9 %	16 %	47 %
r_{eD} median error, p_i unknown	20 %	20 %	30 %
90th percentile	51 %	51 %	73 %
$\ln C_D$ mean absolute error	0.026	0.023	0.042
ring $\ln k$ MAE, $r_D < 100$	0.04–0.09	0.04–0.09	0.08–0.16
ring $\ln k$ MAE, r_D 100–300	0.06	0.06	0.14
ring $\ln k$ MAE, r_D 300–1000	0.15	0.17	0.33
ring $\ln k$ MAE, r_D 1000–3000	0.27	0.34	0.50
no-storage flag accuracy	100 %	100 %	100 %
coverage of $\pm 2\sigma$: rings / C_D / r_{eD}	94 / 92 / 96 %	96 / 97 / 97 %	90 / 92 / 94 %
classical material balance: usable in	75 %	63 %	9 %
its median r_{eD} error	0.15 %	1.3 %	8.0 %
model on the same tests	1.1 %	1.9 %	7.6 %

Table 4: Final (noise-aware) model on the test set. “Usable” classical material balance: p_i known and the build-up stabilized.

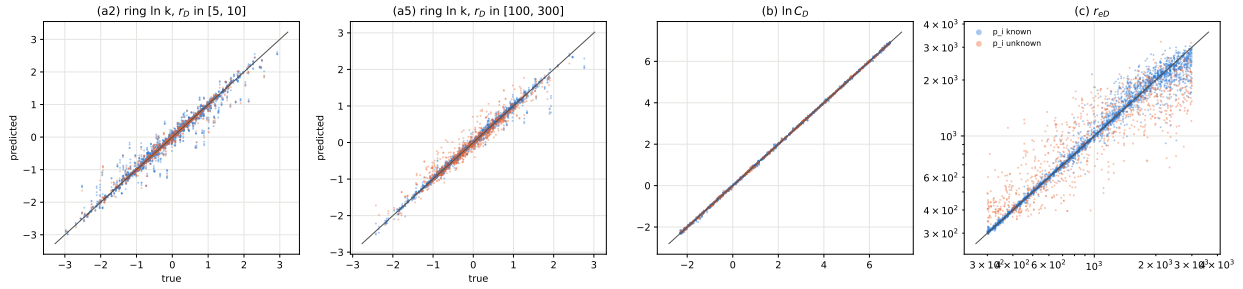


Figure 6: Final model on the test set at 0.1 % gauge noise: predicted vs true ring permeabilities near (r_D 5–10) and far (100–300) from the well, $\ln C_D$, and r_{eD} (blue: p_i known).

above (3.7 % at 0.1 % noise). With little production the depletion $p_i - \bar{p} \approx 2t_p/r_{eD}^2$ is buried in the gauge noise, however long the shut-in. The second is a missing p_i (Section 2.5). The test length matters little once shut-ins last days. The model recognizes these cases: its predicted standard deviation of $\ln r_{eD}$ doubles (0.47 against 0.24) and 82 % of these errors remain within $\pm 2\sigma$. A practical design rule follows: to determine the boundary distance to within about 7 % at 1 % gauge noise with p_i known, produce for $t_p \gtrsim 0.01 r_{eD}^2/k$ in dimensionless time before the shut-in.

7 Discussion

Compared with classical interpretation. In heterogeneous reservoirs the classical plateau permeability depended on the chosen window (median error 23 %) and the semilog skin was misleading; the model returns a radial permeability profile instead. It gives an estimate for every test, including those where the classical specialized methods do not apply (at 1 % noise, classical material balance was usable in 9 % of the tests), it estimates all parameters jointly with calibrated uncertainties (C_D to 2–4 % against about 9 % from the classical unit slope), and it takes milliseconds per test. Where an exact formula applies it remains better: on stabilized tests with a precise gauge and known p_i , material balance gives r_{eD} to 0.15 % against 1.1 % for the model.

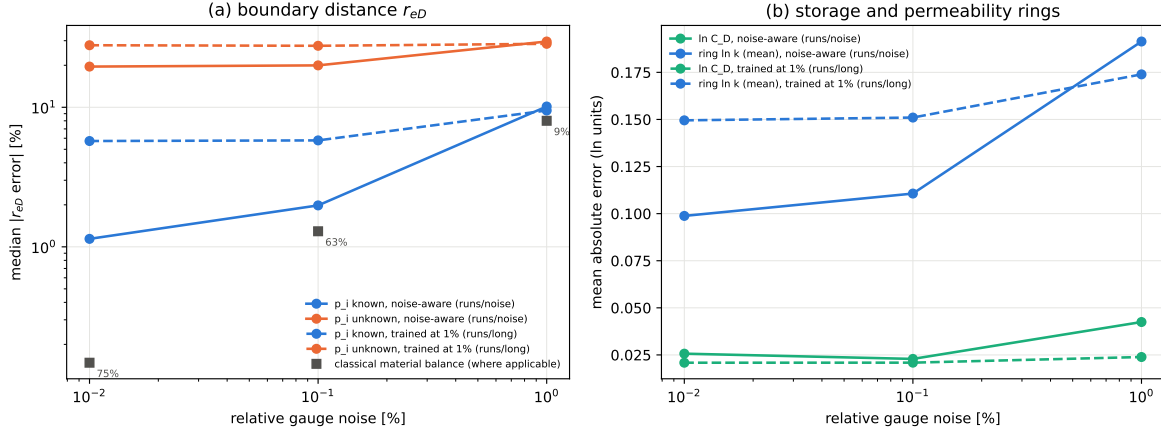


Figure 7: Dependence on the gauge noise. (a) r_{eD} error with and without p_i for the noise-aware model (solid) and a model trained at 1% noise (dashed), and classical material balance where it applies (percentages: share of tests where it is usable). (b) Storage and ring errors.

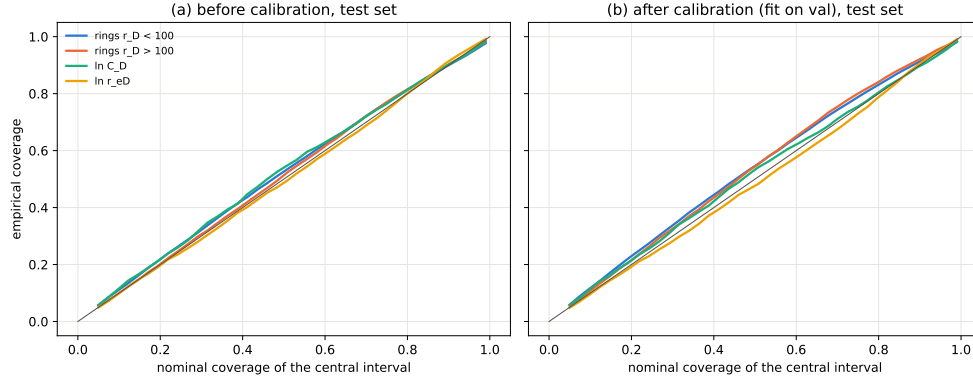


Figure 8: Reliability of the predicted uncertainties on the test set (model trained at 1% noise): empirical vs nominal coverage of central intervals, before and after a validation-fitted scaling. The raw uncertainties are already calibrated.

Limitations. The model is only valid within the physics it was trained on: one well, a closed circular boundary, radial heterogeneity, one constant-rate flow period, single-phase flow. Faults, channels, dual porosity, multiphase effects and rate histories are outside its training, and its uncertainty is not guaranteed to flag such data. Like classical analysis it needs q , B , μ , h , ϕc_t and r_w to make the data dimensionless. It has not yet been tested on field data.

Without the initial pressure. The remaining 20–30% error in r_{eD} without p_i is the volume–permeability ambiguity of Section 2.5. It can be reduced with more data rather than a better network: the flowing-pressure record before shut-in (its pseudo-steady-state slope measures the pore volume without p_i), or two build-ups separated by a known production.

Next steps. Recover the accuracy at 1% noise (longer or wider training); report classical material balance where it applies; add the flowing period as an input; benchmark the network against the model-based inversion on many tests; widen the training physics; test on field build-ups.

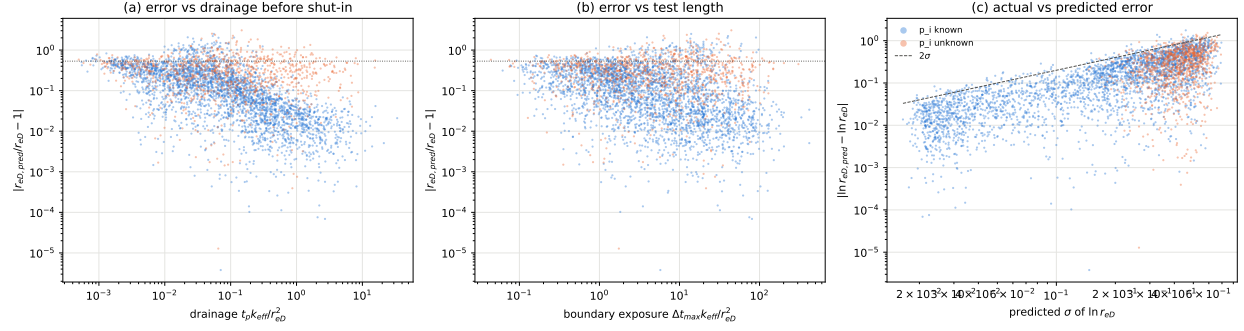


Figure 9: r_{eD} error at 1% noise vs (a) drainage before shut-in, (b) boundary exposure during the test, and (c) the predicted uncertainty (dashed: 2σ).

A Reproducing the results

All code is in `buildup/` (independent of the rest of the repository); tests: `pytest buildup/tests` (52 tests). Main scripts: `solver.py`, `analytic.py` (reference solutions), `wells.py` (build-up tools), `permeability.py`, `dataset.py`, `invert.py`, `examples.py`, `model.py`, `train.py`, `evaluate.py`, `calibrate.py`, `analyze_re_tail.py`; figures by the `fig_*.py` scripts. The plan and the full record of decisions and results are in `buildup/tasks.md`.